

Speech Communication (BTech =>CT 437/ PhD/MTech => CT 540)

Course Title: Speech Communication

Credit Structure: (L-T-P-Cr): 3-0-2-4

Prerequisites (if any): Basic Knowledge of Signals and Systems is desirable but will also be revised in the initial stage of the course.

Course Offered as **Technical Elective**: BTech (ICT+CS for Sem V and Sem VII), MTech (ICT, WCSP, Sem III), MTech (ML as elective), MTech (VLSI => audit course) . MSc(DS, Sem III), Open to PhD

A) Grading Policy

Class Attendance = 15 %

1st In-Sem= 15 %

2nd In-Sem- 15 %

Course Project =15 %

Lab= 10 %

Endsem =30 %

B) **Textbook**: I will be following a world-class textbook in the entire course

Textbook: Thomas F. Quatieri, "Discrete-Time Speech Signal Processing," Pearson Education, 2004.

C) Exam Schedule and Syllabus w.r.t Course Coverage

- 1) Course Project -> Students are encouraged to do projects in diversified areas (not limited to) **Deep Learning/Machine Learning** Applications to Speech Problems (such as Voice Conversion, Automatic Speech Recognition (ASR), Voice Assistants, Assistive Speech Technologies, Language Identification, Speech Enhancement, Speech Synthesis)
- 2) 1st In-Sem -> Chapter 1 (Introduction to Speech Communication), and Chapter 2 (Basics Required for the Course), **Generative Adversarial Networks** (GANs) for Speech Applications
- 3) 2nd In-Sem -> Chapter 3 (Physiology of Speech Production), Chapter 4 (Acoustics of Speech Production)
- 4) EndSem (Entire Syllabus) = More emphasis on => Chapter 5 Linear Prediction (LP) of Speech, Chapter 6 Cepstrum Analysis and MFCC

Note:

- Fundamental knowledge of first six chapters is important perform well in Endsem,

- Chapter numbers above refer to the Quatieri's textbook mentioned above.

Course objective: To study fundamentals of speech production (physiological, acoustical and aeroacoustic perspective), speech analysis methods- STFT, spectrograms, LP analysis, and cepstrum analysis. A case study on Automatic Speech and Speaker Recognition.

Key Laboratory Experiments (Tentative)	
LAB1	Introduction to MATLAB Programming
LAB2	Speech Analysis: Reading Speech Waveform, Speech Framing at suprasegmental (< 4 ms), segmental (10-30 ms), and suprasegmental level (~ 200 ms)
LAB3	Significance of Phase of Fourier Transform: A Case Study for Speech
LAB4	Spectrographic Analysis of Speech, Audio, Music, and Infant Cry Signals Narrowband vs. wideband spectrograms, Displaying Mel Spectrograms
LAB4	Estimation of Instantaneous Frequency (IF) using Hilbert Transform and Relevance to Speech Processing
LAB5	Design of 2 nd order Digital Resonator and its relevance to Speech Formant Estimation and their – 3dB Bandwidths
LAB6	Linear Prediction Analysis of Speech Wave: Estimation of Speech Excitation Source using LP residual, LP Spectrum vs. STFT Spectrum
LAB7	Formant and – 3dB bandwidth Estimation using LP Methods
LAB 8	Cepstrum Analysis of Speech Signals=> Estimation of Voiced vs. Unvoiced Regions, Fundamental Frequency (Fo) using Cepstrum, cepstrally-smoothed spectrum
LAB 9	Estimation Fundamental Frequency (Fo) using Autocorrelation, LP Residual, Hilbert Envelope, etc.
LAB 10	Case Study1: Convolutional Neural Networks (CNN) for Speech Applications
LAB 11	Case Study 2: Web-Scale Supervised Pre-training for Speech Recognition (Whisper)
LAB 12	Case Study 3: Generative Adversarial Networks (GANs) for Speech Enhancement, Voice Conversion, etc.

Recent Success Story: Students who took this elective in the last Autumn 2021-2024 succeeded in getting the following research papers, which will help them for job and/or the higher studies !!

- [1] Hemant A. Patil, Aastha Kachhi, and Ankur T. Patil, "CQT-based cepstral features for classification of normal vs. pathological infant cry," in IEEE Trans. Speech Audio, and Language Processing, vol. 32, pp. 4713-4726, 2024.

- [2] Arth J. Shah, Ravindrakumar M. Purohit, Dharmendra H. Vaghera, and Hemant A. Patil, "MLADDC: Multi-Lingual Audio Deepfake Detection Corpus," In Audio Imagination: NeurIPS (NIPS) Workshop on AI-Driven Speech, Music, and Sound Generation, Vancouver, Canada, , pp. 1-9, 14 Dec. 2024.
- [3] Arth J. Shah, Hiya Chaudhari, and Hemant A. Patil, "Infant Cry Classification Using Modified Group Delay Cepstral Coefficients," in Antonacopoulos, A., Chaudhuri, S., Chellappa, R., Liu, CL., Bhattacharya, S., Pal, U. (Eds.) Int. Conf. Pattern Recognition (ICPR), Lecture Notes in Computer Science (LNCS), Springer, vol. 15314, pp. 275–289, 2024.
- [4] Japan V. Bhatt, Harsh C. Patel, and Hemant A. Patil, "Noise Robust Whisper Features for Dysarthric Automatic Speech Recognition," in The Speaker and Language Recognition Workshop, Speaker Odyssey, Quebec, Canada, 18-21 June 2024. (**The only paper from India !**)
- [5] Siddharth Rathod, Monil Charola, Akshat Vora, Yash Jogi, and Hemant A. Patil, "Whisper Features for Dysarthric Severity-Level Classification," in INTERSPEECH, Dublin, Ireland, 20-24 August 2023.
- [6] Siddharth Rathod, Aastha Kachhi, Priyanka Gupta, and Hemant A. Patil, "Cochlear Filter-based Cepstral Features for Dysarthric Severity-level Classification," in 31st European Signal Processing Conference (EUSIPCO), Helsinki, Finland, Sept. 04-08, 2023.
- [7] Aastha Kachhi, Monil Charola, and Hemant A. Patil, "Whisper Encoder Module based features for Infant Cry Classification," in INTERSPEECH, Dublin, Ireland, 20-24 August 2023.
- [8] Siddharth Rathod, Monil Charola, and Hemant A. Patil, "Noise Robust Whisper Features for Dysarthric Severity-Level Classification," in PReMI, ISI Kolkata, India, LNCS, Springer, Dec. 12-15, 2023.
- [9] Krishna Parmar, Baveet Singh Vora, Shrey Machhar, Hemant A. Patil, Kiran Praveen, and Balaji Radhakrishnan, "Spoken Language Identification using Linear Frequency Residual Cepstral Coefficients," in PReMI, ISI Kolkata, India, LNCS, Springer, Dec. 12-15, 2023

Course Instructor

Prof. Hemant A. Patil

Course Instructor for Speech Communication (CT 437/CT 540),

Associate Editor, IEEE Signal Processing Magazine 2021-2023

ISCA Distinguished Lecturer 2020-2022.

APSIPA Distinguished Lecturer 2018-2019.

Faculty Block-4, DA-IICT Gandhinagar

Tel +91-79-30510-650 Email: hemant_patil@daiict.ac.in

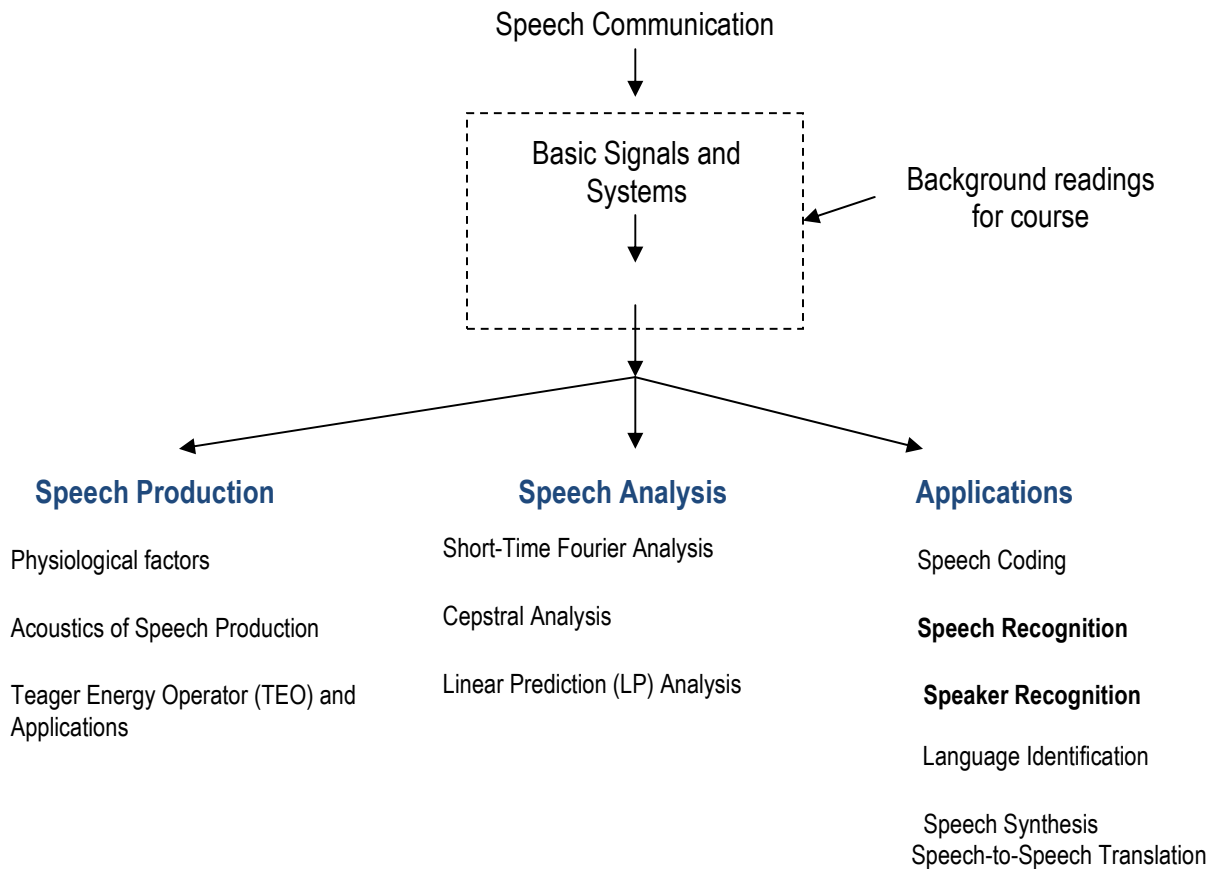


Fig. 1. Activity tree for course on Speech Communication

Course Abstract:- Speech communication is an interdisciplinary subject with different research areas ranging from electrical and communication engineering, linguistics, phonetics, psychology, computer science, etc. Speech communication is the transfer of information from one person to another or from person to machine via speech. This course equips students with different fundamental problems in speech science such as speech production process, speech perception, time-domain and frequency-domain processing, homomorphic speech processing and most popular linear prediction (LP) of speech and finally different diverse speech communication applications such as speech analysis, speech coding, speech synthesis, speech recognition and speaker recognition are discussed. The course is designed in such a way that relationships between different parts of speech communication can be understood easily. In addition to this, cohesive discussion on speech production and speech perception will help audience to understand speech communication process better.

Course Outline: - Fig. 1 shows the activity tree for the course. The course will first review very briefly the basic concepts of DSP and random processes followed by detailed discussion on speech production process from the view point of physiological aspects to acoustic theory of speech production. The unique feature of the course will be a detailed discussion on aeroacoustic theory of speech production and development of Teager Energy Operator (TEO) and recently proposed variable length TEO (VTEO).